4.7.58

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Question

Find the distance between the parallel lines $3x - 4y + 7 = 0$ and $3x - 4y + 5 = 0$.

Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \tag{1}$$

The given lines can be expressed in matrix form as

$$\begin{pmatrix} 3 & -4 \end{pmatrix} \mathbf{x} = -7 \quad \text{and} \quad \begin{pmatrix} 3 & -4 \end{pmatrix} \mathbf{x} = -5 \tag{2}$$

Here, the normal vector is

$$\mathbf{n} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}, \quad c_1 = -7, \quad c_2 = -5 \tag{3}$$

Solution

The distance between two parallel lines is

$$d = \frac{|c_1 - c_2|}{\|\mathbf{n}\|} \tag{4}$$

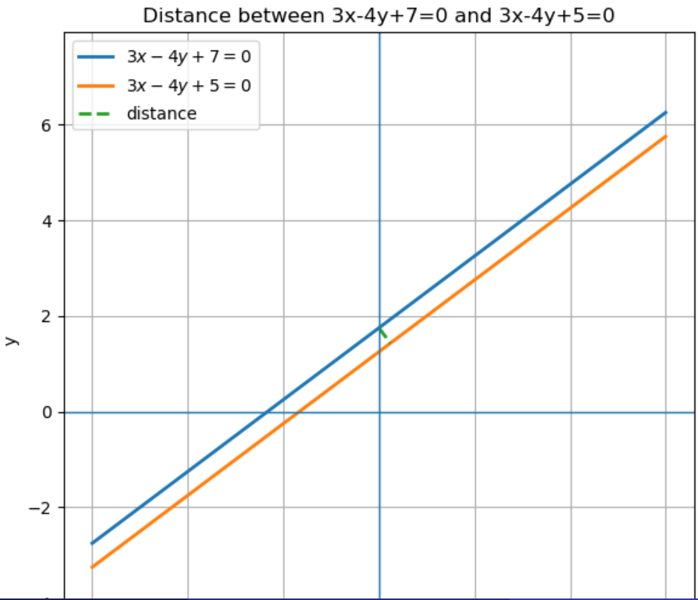
$$\|\mathbf{n}\| = \sqrt{3^2 + (-4)^2} = \sqrt{9 + 16} = 5 \tag{5}$$

$$|c_1 - c_2| = |-7 - (-5)| = |-2| = 2 \tag{6}$$

Therefore,

$$d = \frac{2}{5} \tag{7}$$

Graph


Distance between 3x-4y+7=0 and 3x-4y+5=0
$3x - 4y + 7 = 0$
$3x - 4y + 5 = 0$
distance
6
4
2
0
−2
y

C Code

```c
#include <math.h>

typedef struct {
    double distance; // |c1 - c2| / sqrt(a^2 + b^2)
    double norm_n; // sqrt(a^2 + b^2)
    double delta_c; // |c1 - c2|
} DistOut;

// Distance between parallel lines: a x + b y + c1 = 0 and a x +
    b y + c2 = 0
DistOut distance_parallel_pack(double a, double b, double c1,
    double c2) {
    DistOut out;
    out.norm_n = sqrt(a*a + b*b);
    out.delta_c = fabs(c1 - c2);
    out.distance = (out.norm_n > 0.0) ? (out.delta_c / out.norm_n
        ) : NAN;
```

C Code

```
        return out;
}


    // Convenience for your lines: 3x - 4y + 7 = 0 and 3x - 4y +
        5 = 0
DistOut example_3m4_c7_c5(void) {
    return distance_parallel_pack(3.0, -4.0, 7.0, 5.0);
}
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Lines: a x + b y + c1 = 0 and a x + b y + c2 = 0
a, b = 3.0, -4.0
c1, c2 = 7.0, 5.0

# Distance formula (same as in C)
norm_n = (a*a + b*b)**0.5
dist = abs(c1 - c2) / norm_n
print(fDistance = {dist:.6g}) # should be 2/5 = 0.4

# Solve y = (-a x - c)/b for plotting (b  0 here)
xs = np.linspace(-6, 6, 400)
y1 = (-a*xs - c1)/b
y2 = (-a*xs - c2)/b
```

Python Code

```
   plt.figure(figsize=(6, 6))
plt.plot(xs, y1, linewidth=2, label=r$3x-4y+7=0$)
plt.plot(xs, y2, linewidth=2, label=r$3x-4y+5=0$)

# Draw a perpendicular segment between the lines through an easy
    x (x=0)
x0 = 0.0
y_on_L1 = (-a*x0 - c1)/b
# Perpendicular direction is along n = (a,b); unit vector:
ux, uy = a/norm_n, b/norm_n
# End point on L2 is at distance 'dist' along (ux,uy):
x1p = x0 + dist*ux
y1p = y_on_L1 + dist*uy
plt.plot([x0, x1p], [y_on_L1, y1p], linestyle=--, linewidth=2,
    label=distance)
```

```
# Decorations
plt.axhline(0, linewidth=1)
plt.axvline(0, linewidth=1)
plt.axis(equal)
plt.grid(True)
plt.legend()
plt.xlabel(x)
plt.ylabel(y)
plt.title(Distance between 3x-4y+7=0 and 3x-4y+5=0)
plt.tight_layout()
plt.show()
```

callc.py

```
import ctypes
from ctypes import c_double, Structure

# Adjust filename for macOS: ./libdistlines.dylib
lib = ctypes.CDLL(./libdistlines.so)

class DistOut(Structure):
    _fields_ = [(distance, c_double),
               (norm_n, c_double),
               (delta_c, c_double)]

lib.distance_parallel_pack.argtypes = [c_double, c_double,
    c_double, c_double]
lib.distance_parallel_pack.restype = DistOut

lib.example_3m4_c7_c5.argtypes = []
lib.example_3m4_c7_c5.restype = DistOut
```

```
# Example for your lines 3x - 4y + 7 = 0 and 3x - 4y + 5 = 0
res = lib.example_3m4_c7_c5()
print(Lines: 3x - 4y + 7 = 0 and 3x - 4y + 5 = 0)
print(fn = {res.norm_n:.6g}, |c| = {res.delta_c:.6g})
print(fDistance = {res.distance:.6g})

# General call (uncomment to try other lines with same a,b):
# g = lib.distance_parallel_pack(3.0, -4.0, 7.0, 5.0)
# print(g.distance)
```